

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

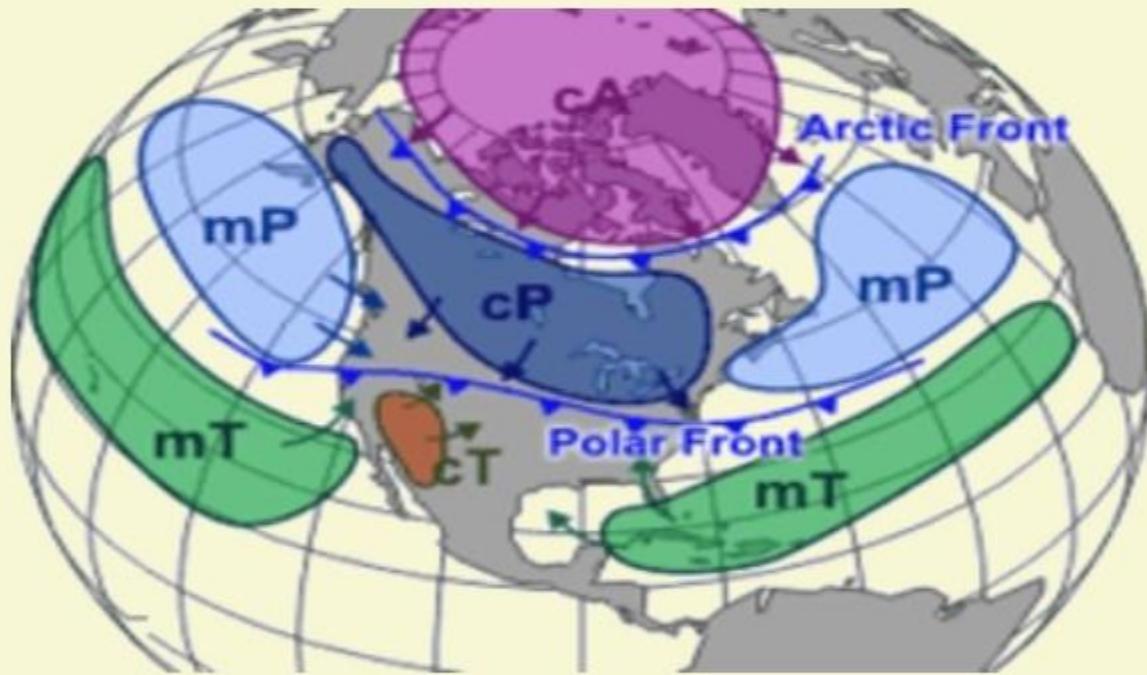
MODULE – 22

- **AIR MASSES AND FRONTS**
- **TEMPERATE CYCLONES**

CLIMATOLOGY MODULES

Atmosphere	▪ Composition and structure; weather and climate;
Insolation	▪ Factors controlling insolation; heat budget of the earth — heating and cooling of atmosphere (conduction, convection, terrestrial radiation, advection);
Temperature	▪ Factors controlling temperature; distribution of temperature — horizontal and vertical; inversion of temperature;
Pressure	▪ Pressure belts; winds-planetary, seasonal and local; air masses and fronts; tropical and extra-tropical cyclones;
Precipitation	▪ Evaporation; condensation — dew, frost, fog, mist and cloud; rainfall — types and world distribution;
World Climate and Climatic Changes	▪ Different types of Climate

AIR MASSES



- 1 When the air remains over a homogenous area for a sufficiently longer time, it acquires the characteristics of the area.
- 2 The homogenous regions can be the vast ocean surface or vast continents.
- 3 The air with distinctive characteristics in terms of temperature and humidity is called an airmass.
- 4 Airmass is defined as a large body of air having little horizontal variation in temperature and moisture. The homogenous surfaces, over which air masses form, are called the **source regions**.

AIR MASSES

5 The air masses are **classified according to the source regions**. There are **five major source regions**.

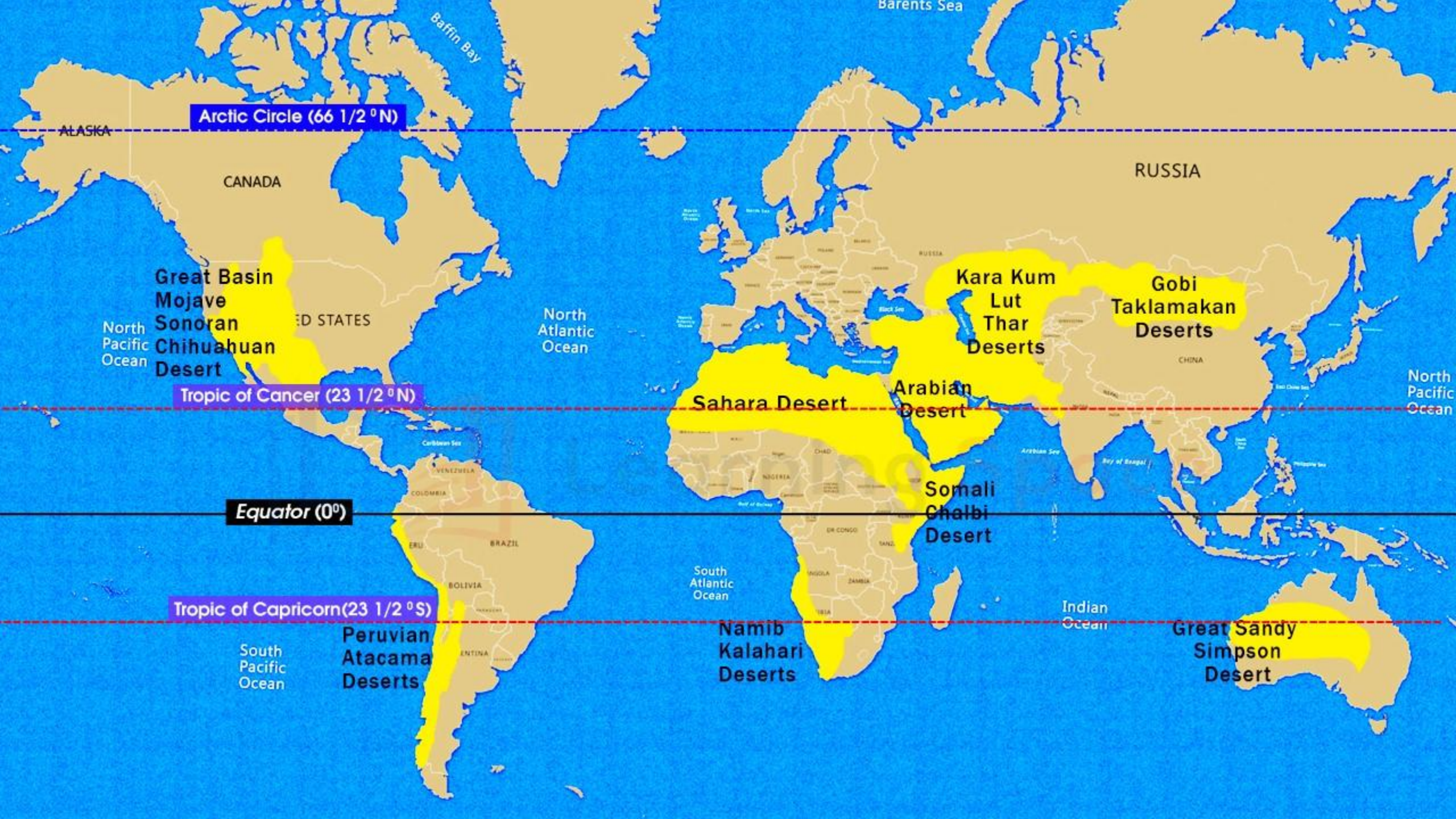
6 These are:

- I. Warm tropical and subtropical oceans,
- II. The subtropical hot deserts
- III. The relatively cold high latitude oceans
- IV. The very cold snow covered continents in high latitudes
- V. Permanently ice covered continents in the Arctic and Antarctica.

7 Accordingly, following types of air-masses are recognised

- I. Maritime tropical (mT)
- II. Continental Tropical (cT)
- III. Maritime Polar (mP)
- IV. Continental Polar (cP)
- V. Continental Arctic (cA).

Tropical air masses are warm and polar air masses are cold.



Arctic Circle (66 1/2 °N)

ALASKA

CANADA

Great Basin
Mojave
Sonoran
Chihuahuan
Desert

North
Pacific
Ocean

UNITED STATES

North
Atlantic
Ocean

Tropic of Cancer (23 1/2 °N)

Equator (0°)

Tropic of Capricorn (23 1/2 °S)

South
Pacific
Ocean

Peruvian
Atacama
Deserts

Sahara Desert

Arabian
Desert

Kara Kum
Lut
Thar
Deserts

Gobi
Taklamakan
Deserts

Somali
Chalbi
Desert

Namib
Kalahari
Deserts

Great Sandy
Simpson
Desert

RUSSIA

CHINA

North
Pacific
Ocean

Indian
Ocean

AIR MASSES

UPSC 2016 GS-1 QUESTION

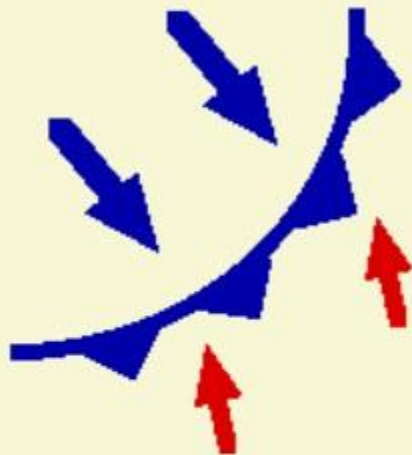
Discuss the concept of air mass and explain its role in macro-climatic changes.
(12.5 marks) (200 words)

HINTS

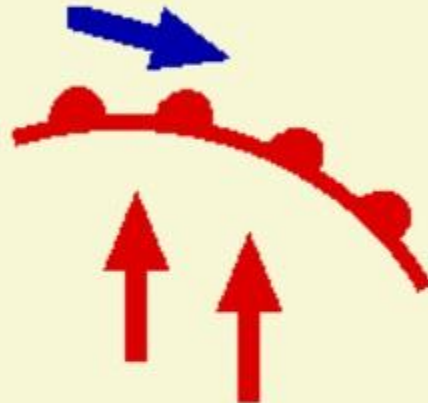
- 1 Define airmass and give a description of how it affects the area it invades.
- 2 Creation of **frontal cyclonic circulation** by the mixing of contrasting airmasses
- 3 **Continental Tropical airmass**- generation of heat waves. Eg: Orissa, AP; aridity of Sahel region in Africa.
- 4 **Continental Polar airmasses** from N.Canada can migrate and bring in frigid winters to as far as Central American regions
- 5 South-West monsoons of India influenced by **tropical maritime airmasses** from Indian Ocean
- 6 Generation of extreme patterns like cyclones, heavy precipitation, heat waves, cold waves affects the flora, fauna and topography of a region.

FRONTS

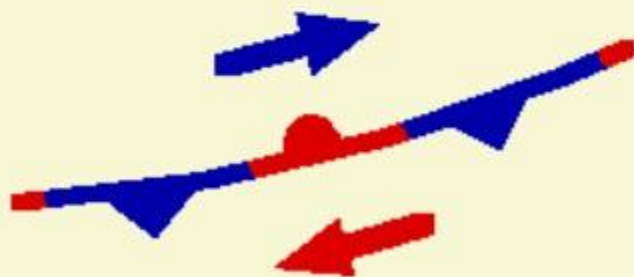
Cold Front



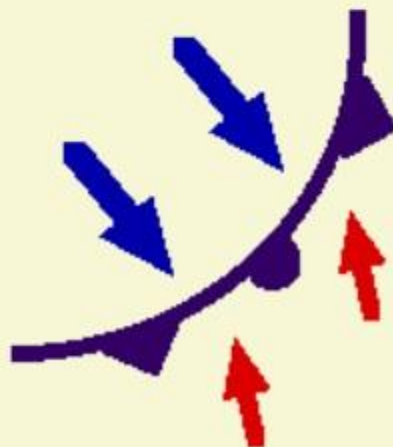
Warm Front



Stationary Front



Occluded Front



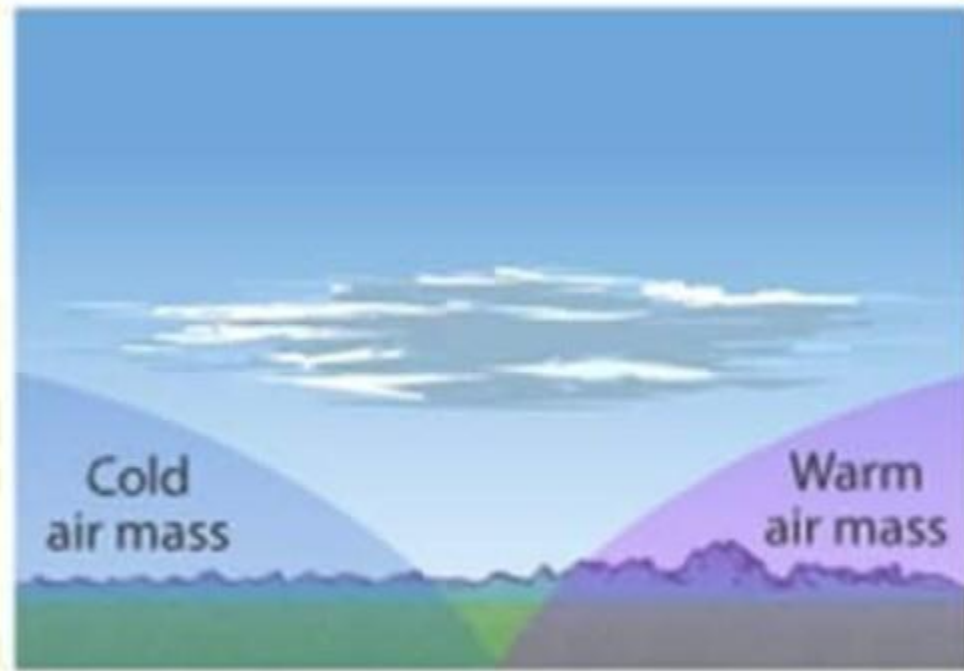
1 When two different air masses meet, the boundary zone between them is called a front.

2 The process of formation of the fronts is known as frontogenesis.

3 There are four types of fronts:

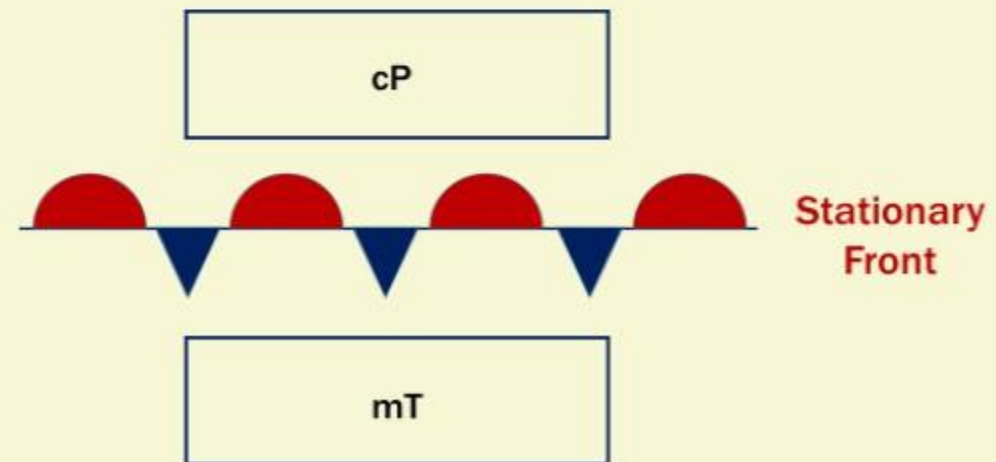
- Cold
- Warm
- Stationary
- Occluded

FRONTS

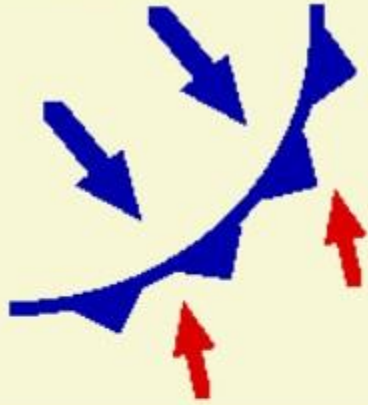


Stationary front

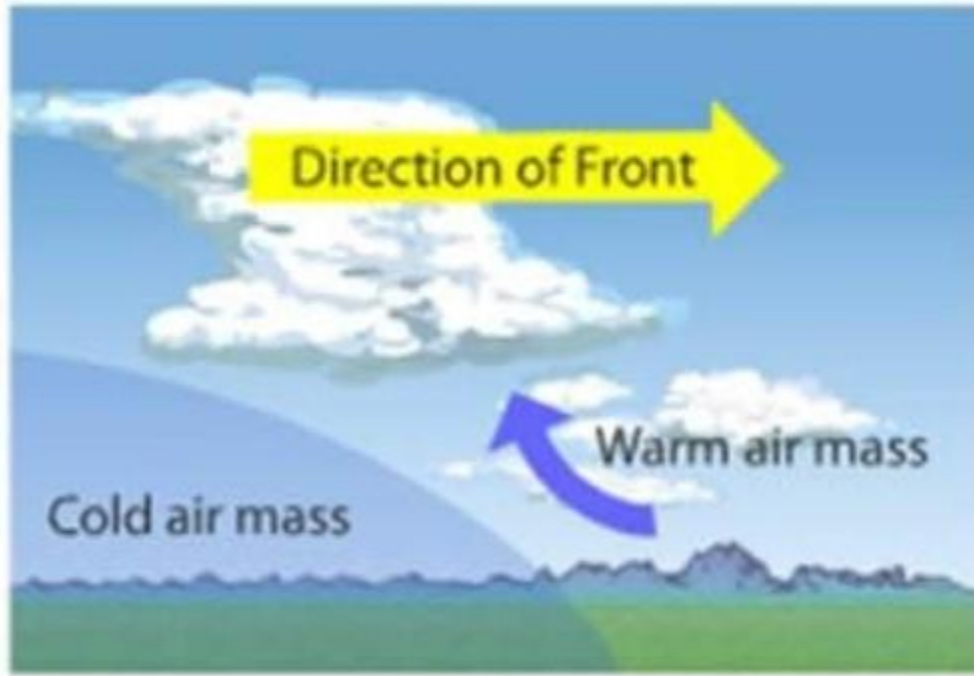
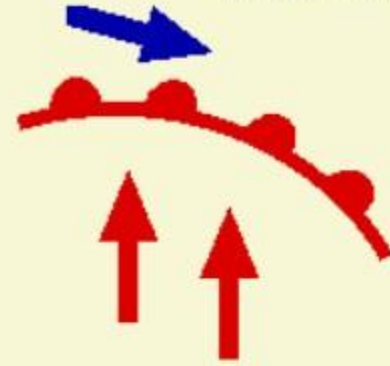
- 4 When the front remains stationary, it is called a **stationary front**.
- 5 When the cold air moves towards the warm air masses, its contact zone is called the **cold front**, whereas if the warm air mass moves towards the cold air mass, the contact zone is a **warm front**.



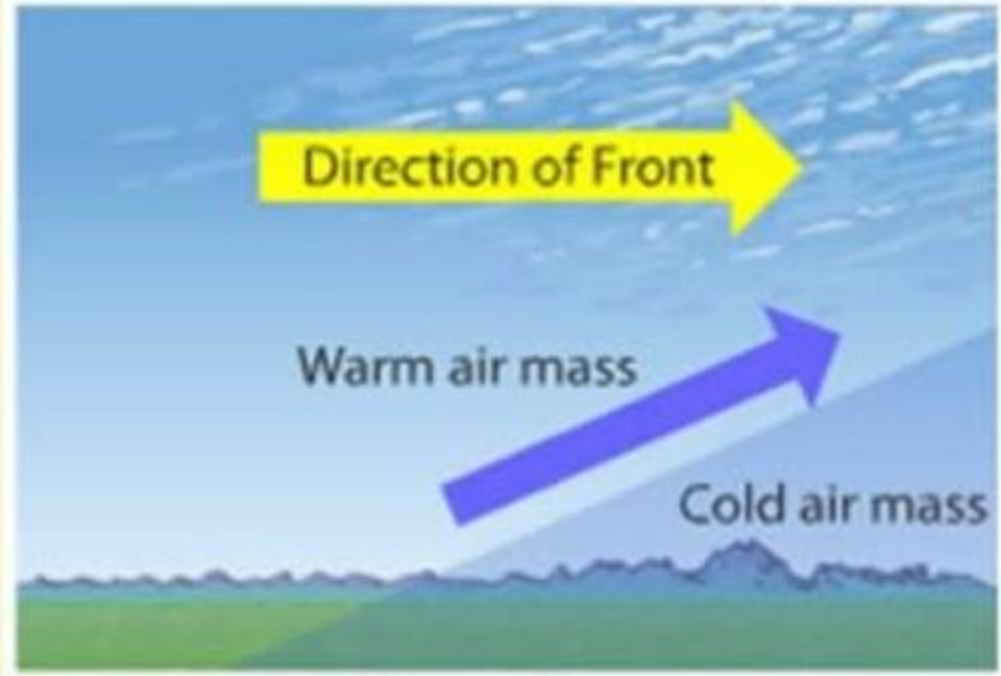
Cold Front



Warm Front

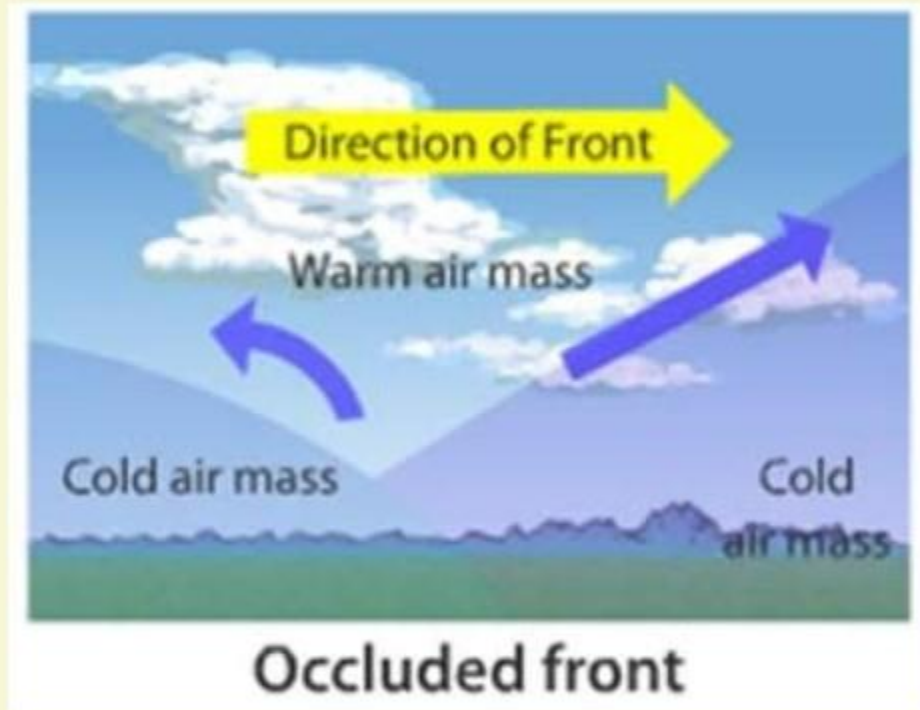


Cold front



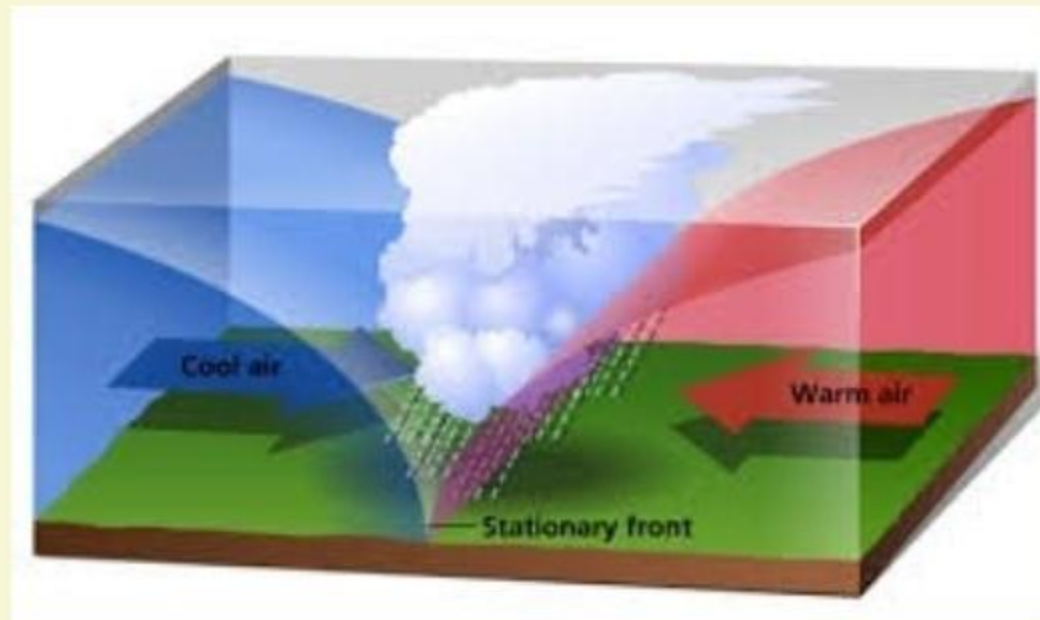
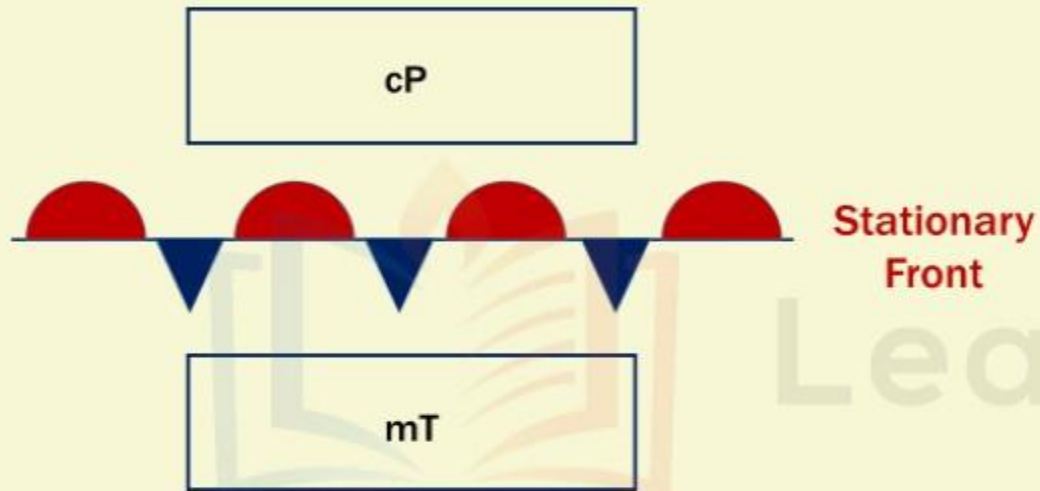
Warm front

FRONTS



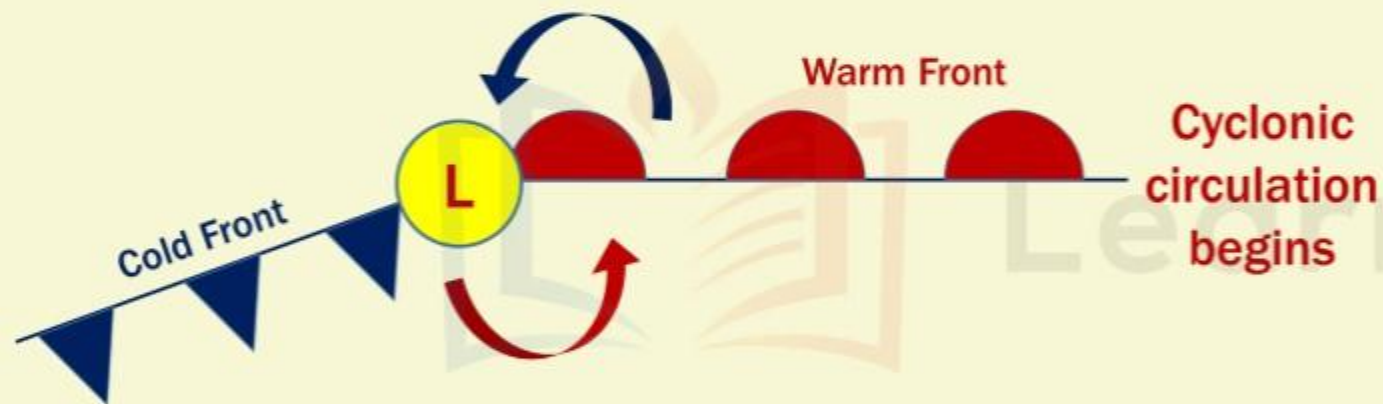
- 6** If an air mass is fully lifted above the land surface, it is called Occluded front.
- 7** The fronts are characterised by steep gradient in temperature and pressure.
- 8** They bring abrupt changes in temperature and cause the air to rise to form clouds and cause precipitation.

EXTRA TROPICAL CYCLONES



- 1** The systems **developing in the mid and high latitudes** beyond the tropics are called the Middle Latitude or Extra tropical cyclones
- 2** The passage of front causes abrupt changes in the weather conditions over the area in the middle and high latitudes.
- 3** Extra tropical cyclones form along the polar front. Initially, the front is stationary.

EXTRA TROPICAL CYCLONES

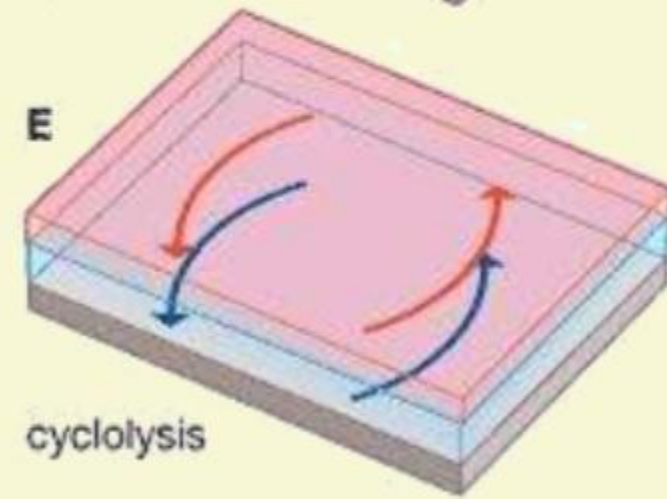
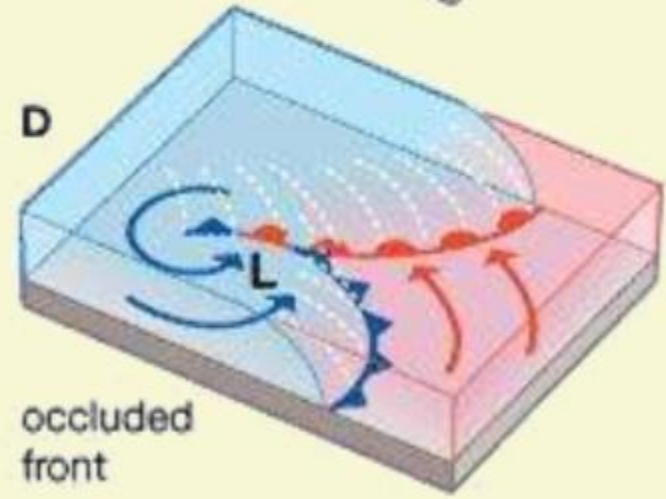
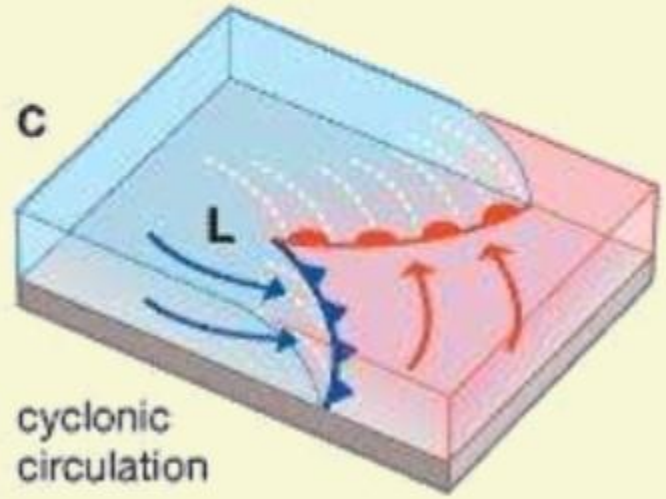
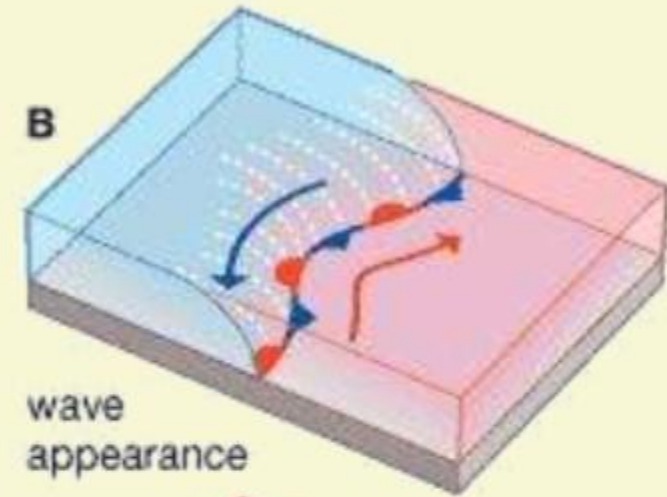
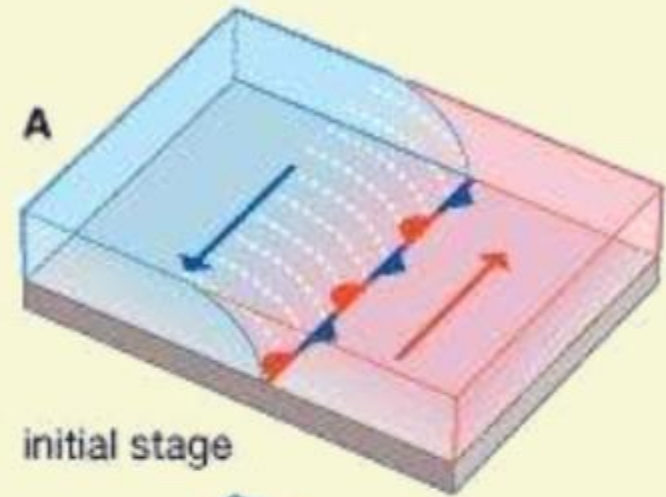


4 In the northern hemisphere, warm air blows from the south and cold air from the north of the front.

5 When the pressure drops along the front, the warm air moves northwards and the cold air move towards south **setting in motion an anticlockwise cyclonic circulation.**

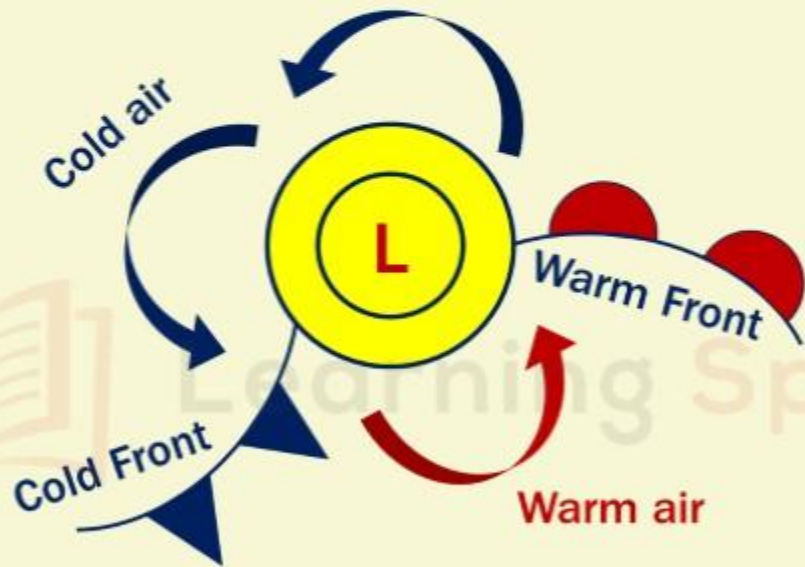
Cyclogenesis

-  Cold air
-  Cold-air flow
-  Warm air
-  Warm-air flow



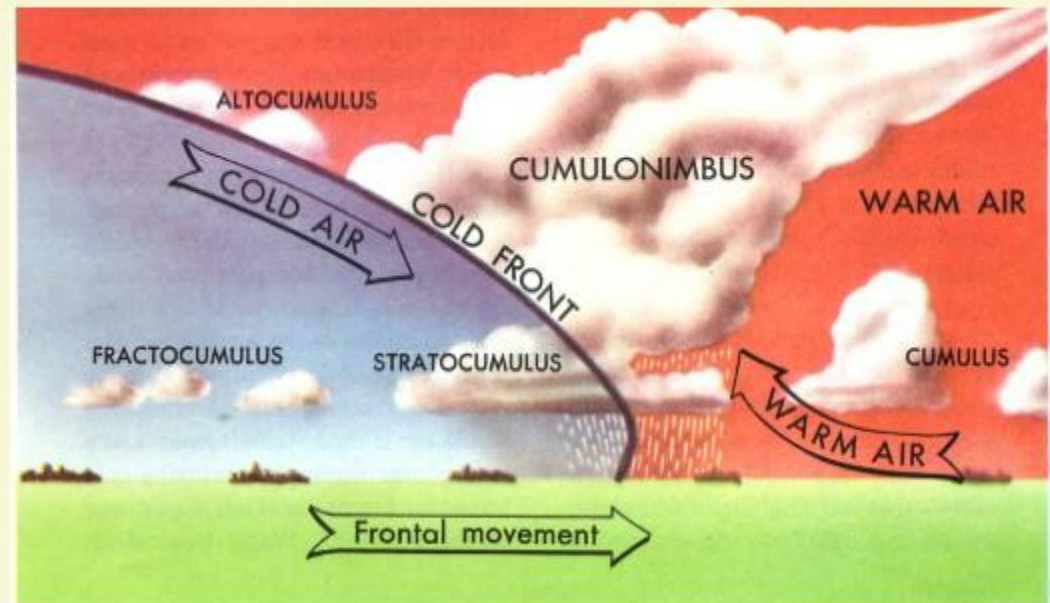
EXTRA TROPICAL CYCLONES

6 The cyclonic circulation leads to a well developed extra tropical cyclone, with a warm front and a cold front. The plan and cross section of a well developed cyclone is given.

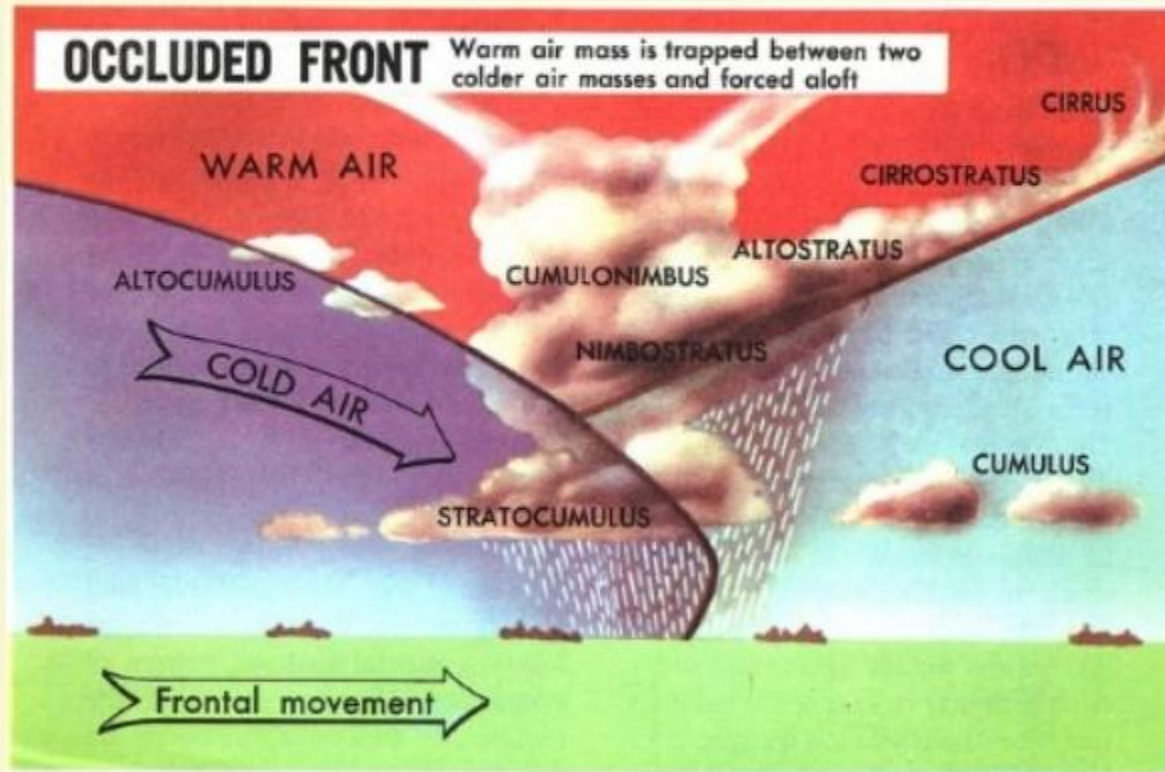


**Well – developed
temperate cyclone**

7 The warm air glides over the cold air and a sequence of clouds appear over the sky ahead of the warm front and cause precipitation.

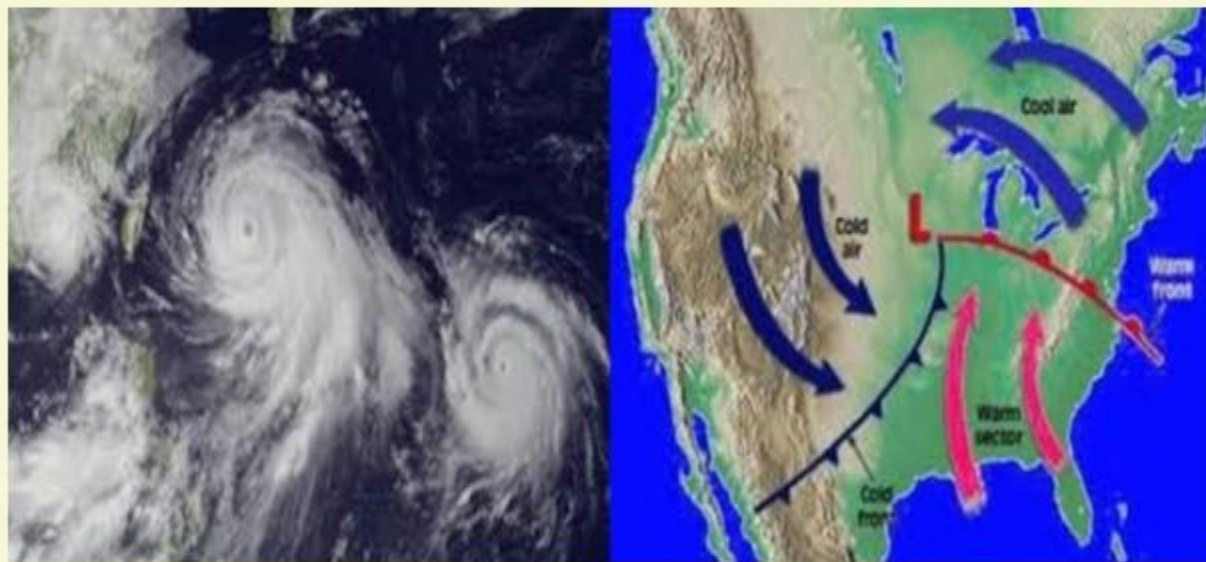


EXTRA TROPICAL CYCLONES



- 8** The cold front moves faster than the warm front ultimately overtaking the warm front. The warm air is completely lifted up and the front is occluded and the cyclone dissipates.
- 9** The processes of wind circulation both at the surface and aloft are closely interlinked. The extra tropical cyclone differs from the tropical cyclone in number of ways. The extra topical cyclones have a clear frontal system, which is not present in the tropical cyclones.

EXTRA TROPICAL CYCLONES



10 They cover a larger area and can originate over the land and sea. Whereas tropical cyclones originate only over the seas and on reaching the land they dissipate.

11 The extra tropical cyclone affects a much larger area as compared to the tropical cyclone but the wind velocity in tropical cyclone is much higher and it is thus more destructive.

12 The extra tropical cyclones move from west to east but tropical cyclone, move from east to west.

10

UPSC CIVIL SERVICES
PRELIMS - 2018

MCQs DISCUSSION

GEOGRAPHY

SESSION - 10

30th May, 2018

S.No	TROPICAL CYCLONES	TEMPERATE CYCLONES
1	Smaller in size.	Larger in size and cover a large area.
2	Can originate only overseas and dissipate on reaching the land.	Can originate both over land and sea.
3	No frontal system found here.	Presence of frontal system with a cold and warm font.
4	Wind velocity is much higher attaining up to 400 Kmph are more destructive.	Wind velocity is less 18-32 Kmph.
5	Move from East to West as they are under the influence of trades.	Move from West to East as they are under the influence of westerlies.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 - 2 4 4 4 7 2

0 9 8 4 9 9 4 2 2 9 9